Polyphase Systems

EEE203

Ref- Fundamentals of Electric Circuits (5th edition) Chapter-12

Ref- Introductory Circuit Analysis (12th edition) Chapter-22

Single Phase

- A single-phase ac power system consists of a generator connected through a pair of wires to a load.
- A single-phase three wire system contains two identical sources (equal magnitude and the same phase) that are connected to two loads by two outer wires and the neutral.

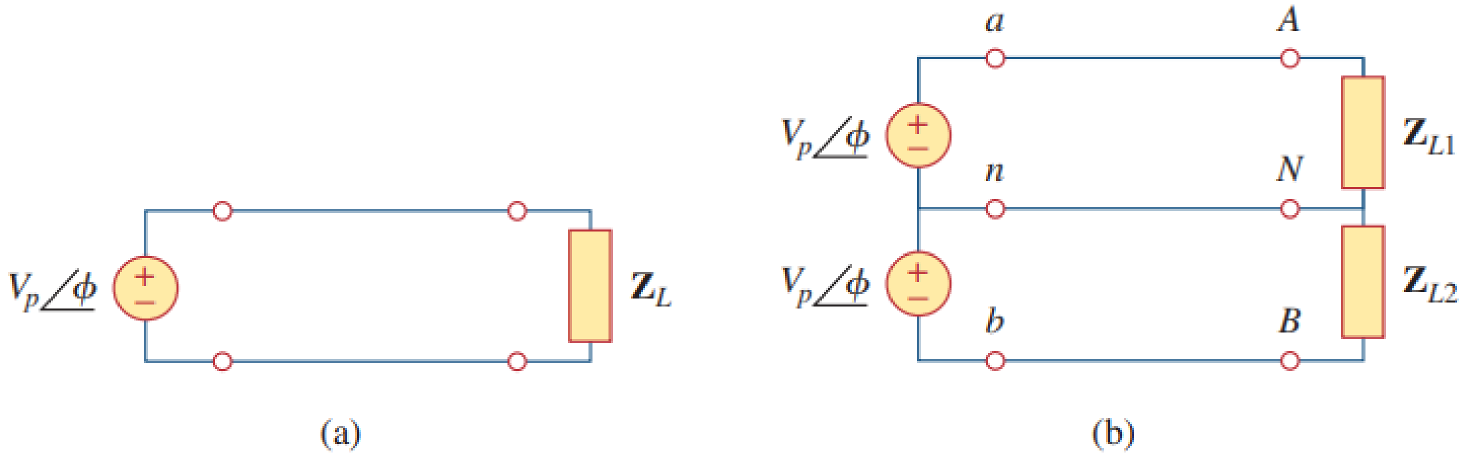


Figure 12.1
Single-phase systems: (a) two-wire type, (b) three-wire type.

Polyphase

- Circuits or systems in which the ac sources operate at the same frequency but different phases are known as polyphase.

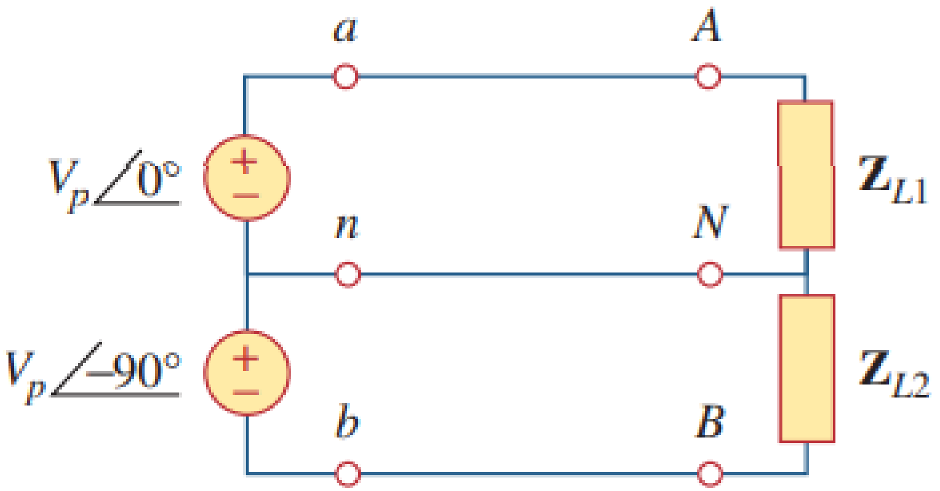


Figure 12.2
Two-phase three-wire system.

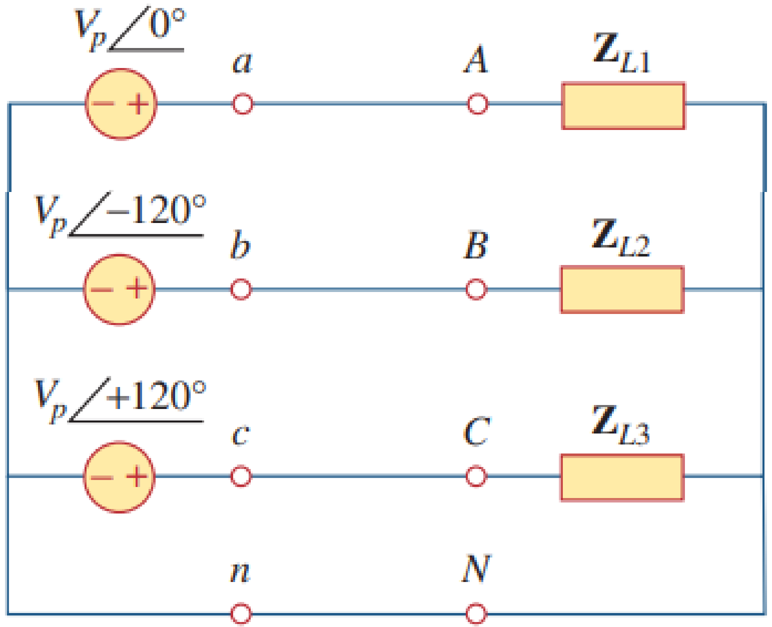


Figure 12.3
Three-phase four-wire system.

Three-Phase System

- Number of phases can be greater than 3.
- Since the three-phase system is by far the most prevalent and most economical polyphase system, discussion is mainly on three-phase systems.

Advantages Associated with a Three-Phase Power Distribution over a Single-Phase System:

1. Thinner conductors can be used to transmit the same kVA at the same voltage, which reduces the amount of copper required (typically about 25% less) and in turn reduces construction and maintenance costs.
2. The lighter power lines are easier to install, and the supporting structures can be less massive and farther apart.
3. And others.

Three-Phase System

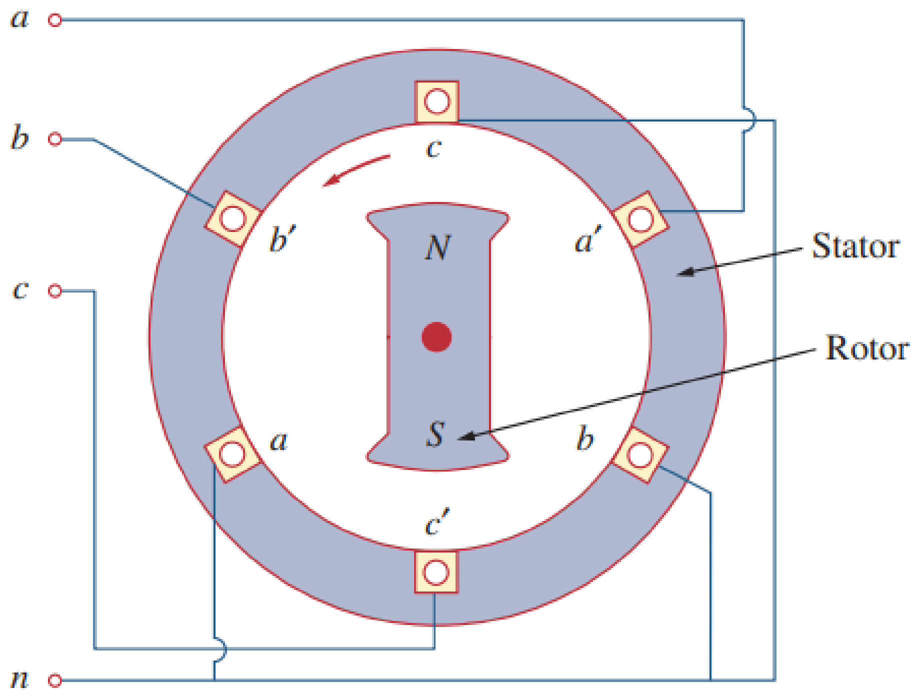


Figure 12.4
A three-phase generator.

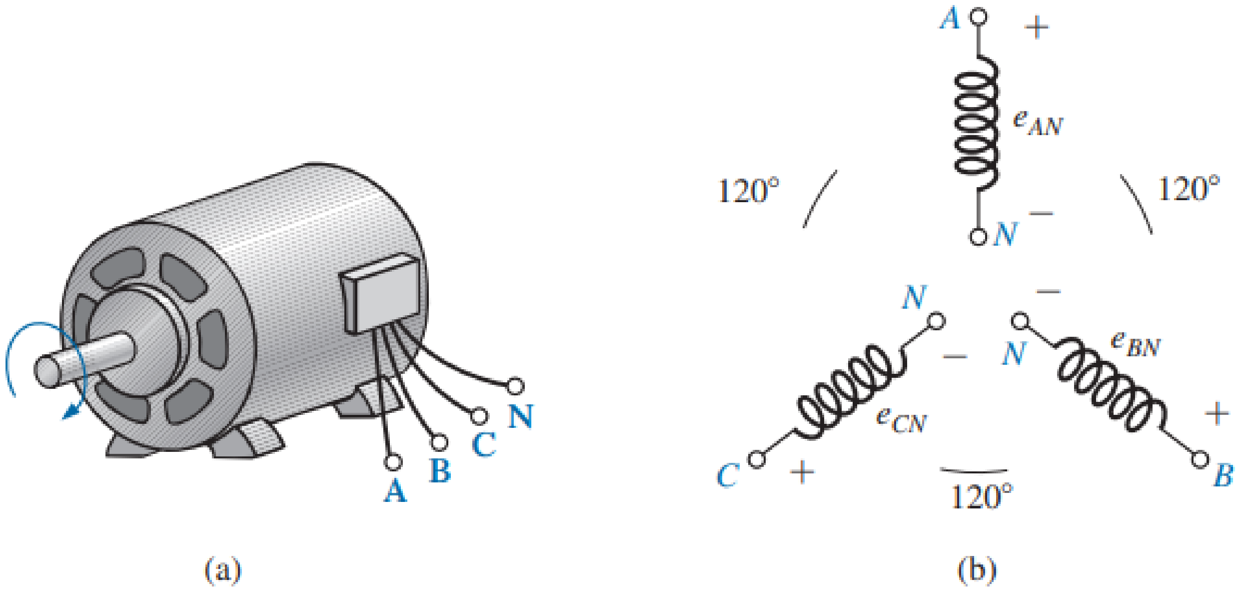


FIG. 24.1
(a) Three-phase generator; (b) induced voltages of a three-phase generator.

Three-Phase System

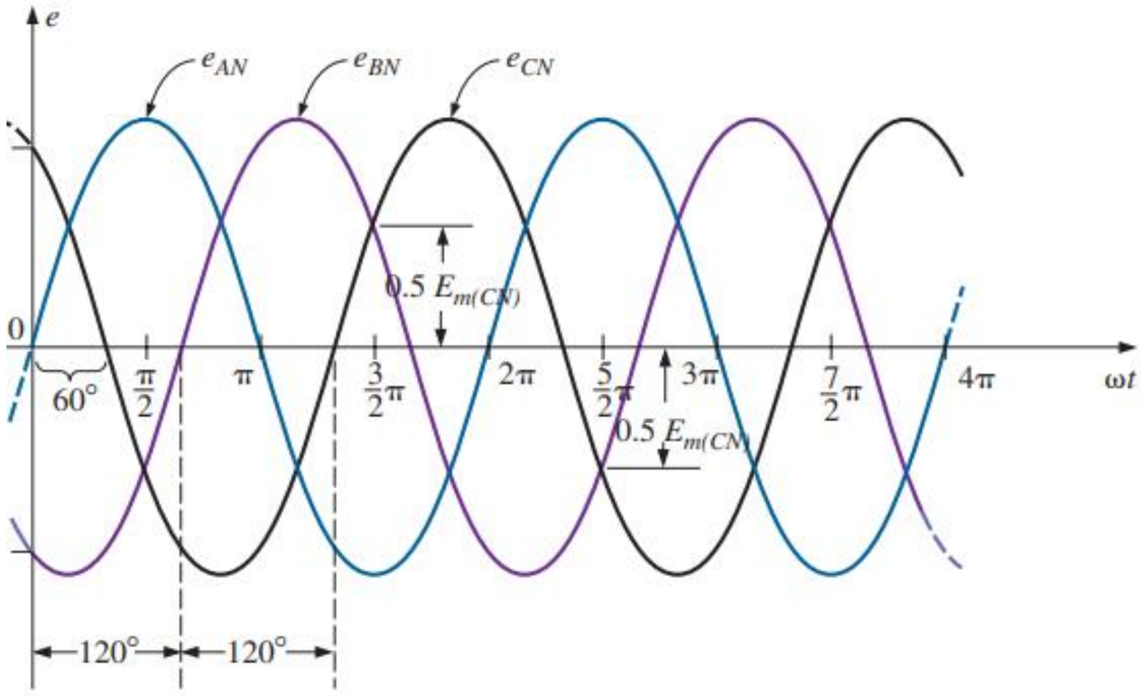


FIG. 24.2
Phase voltages of a three-phase generator.

Time Domain

$$e_{AN} = E_{m(AN)} \sin \omega t$$
$$e_{BN} = E_{m(BN)} \sin(\omega t - 120°),$$
$$e_{CN} = E_{m(CN)} \sin(\omega t - 240°) = E_{m(CN)} \sin(\omega t + 120°)$$

Three-Phase System

Time Domain

$$e_{AN} = E_{m(AN)} \sin \omega t$$

$$e_{BN} = E_{m(BN)} \sin(\omega t - 120°),$$

$$e_{CN} = E_{m(CN)} \sin(\omega t - 240°) = E_{m(CN)} \sin(\omega t + 120°)$$

Phasor Domain

$$\mathbf{E}_{AN} = E_{AN} \angle 0°$$

$$\mathbf{E}_{BN} = E_{BN} \angle -120°$$

$$\mathbf{E}_{CN} = E_{CN} \angle +120°$$

RMS Values

$$E_{AN} = 0.707 E_{m(AN)}$$

$$E_{BN} = 0.707 E_{m(BN)}$$

$$E_{CN} = 0.707 E_{m(CN)}$$

Three-Phase System

Phasor Domain

$$\mathbf{E}_{AN} = E_{AN} \angle 0^\circ$$
$$\mathbf{E}_{BN} = E_{BN} \angle -120^\circ$$
$$\mathbf{E}_{CN} = E_{CN} \angle +120^\circ$$

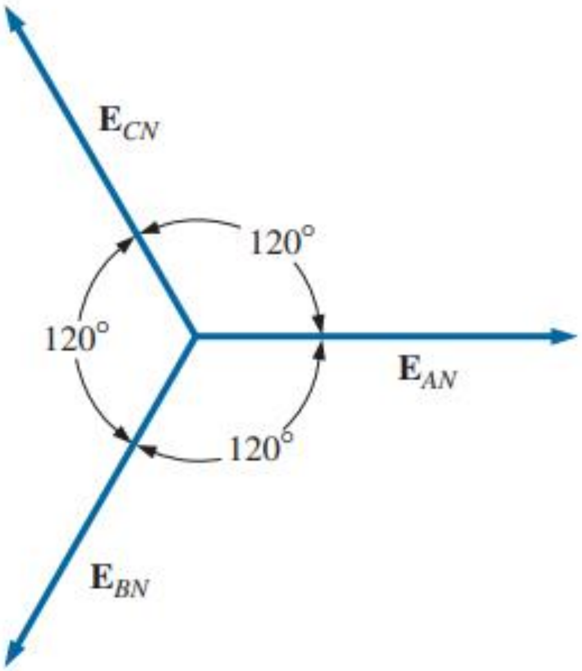
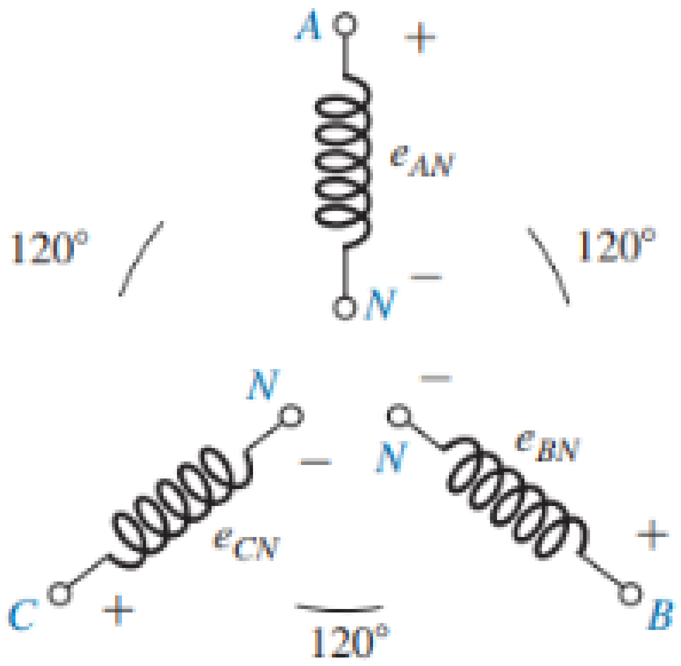


FIG. 24.3
Phasor diagram for the phase voltages of a three-phase generator.

Wye Connection



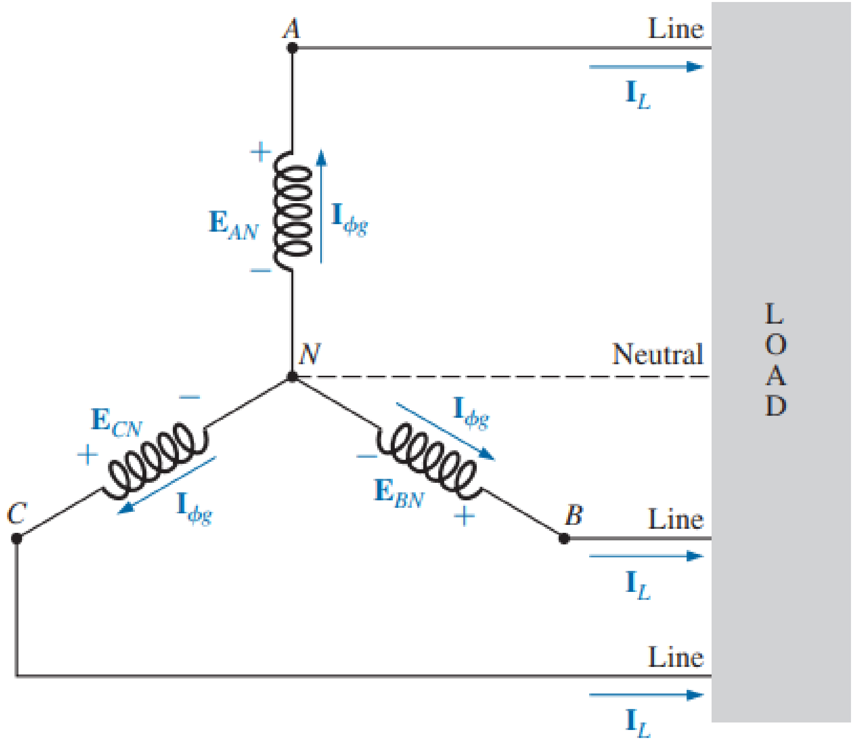


FIG. 24.4
Y-connected generator.

Delta Connection

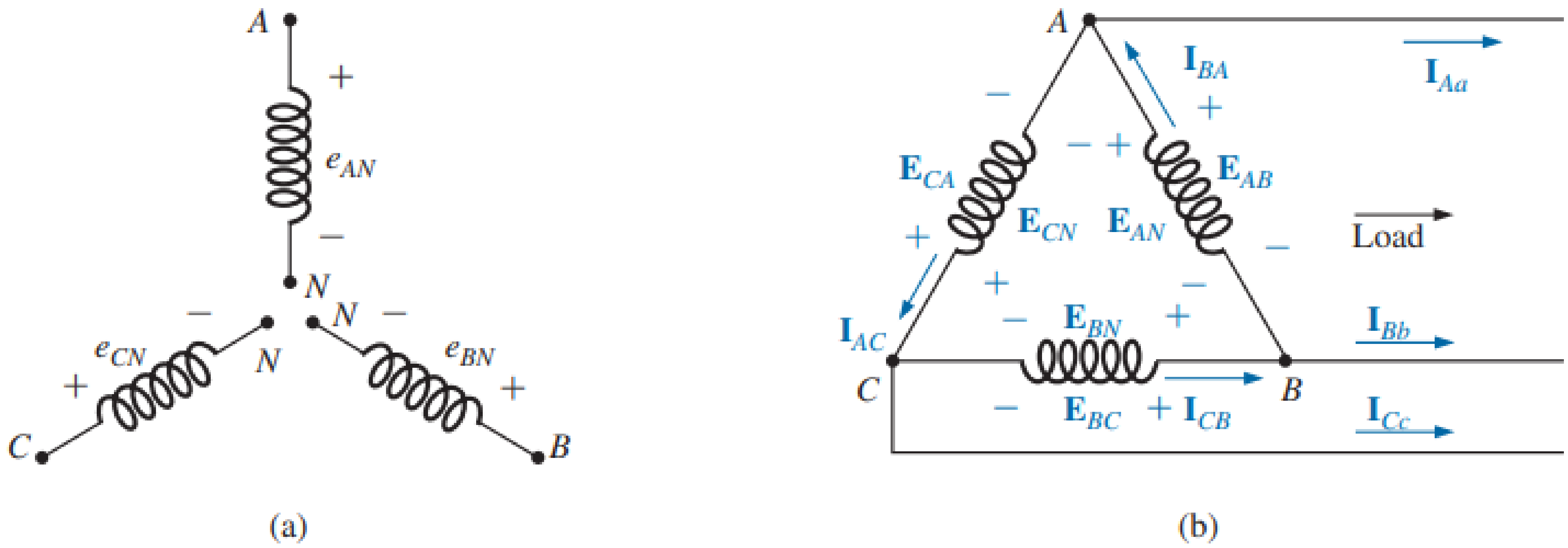


FIG. 24.14
Δ-connected generator.

Note: No neutral connection in delta

Three-Phase System

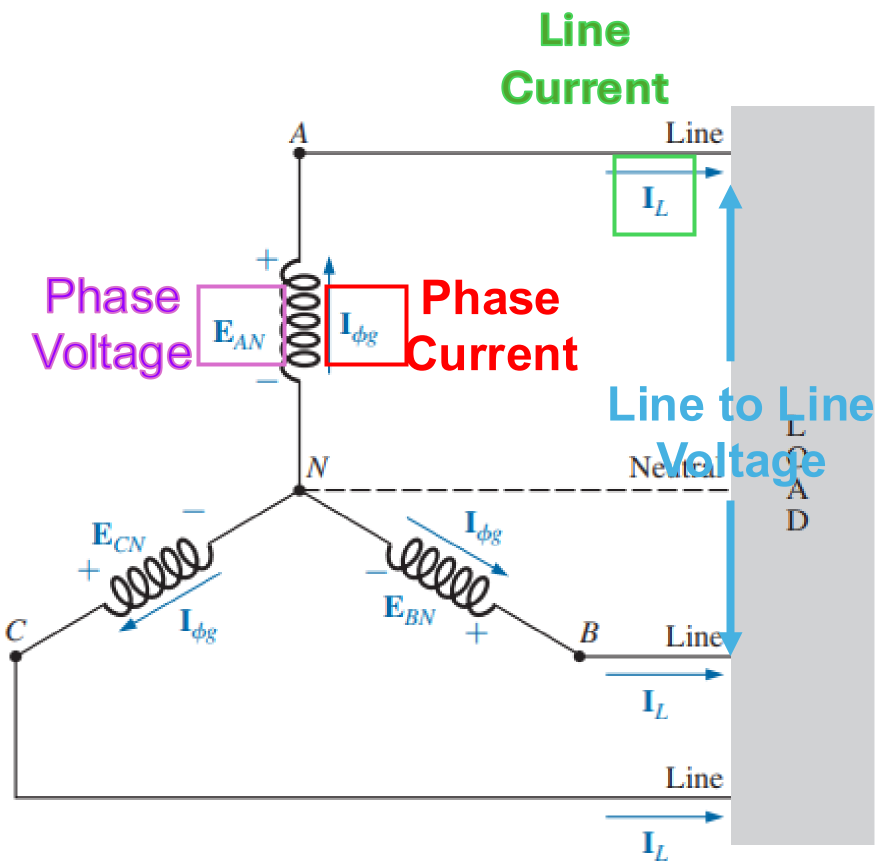


FIG. 24.4
Y-connected generator.

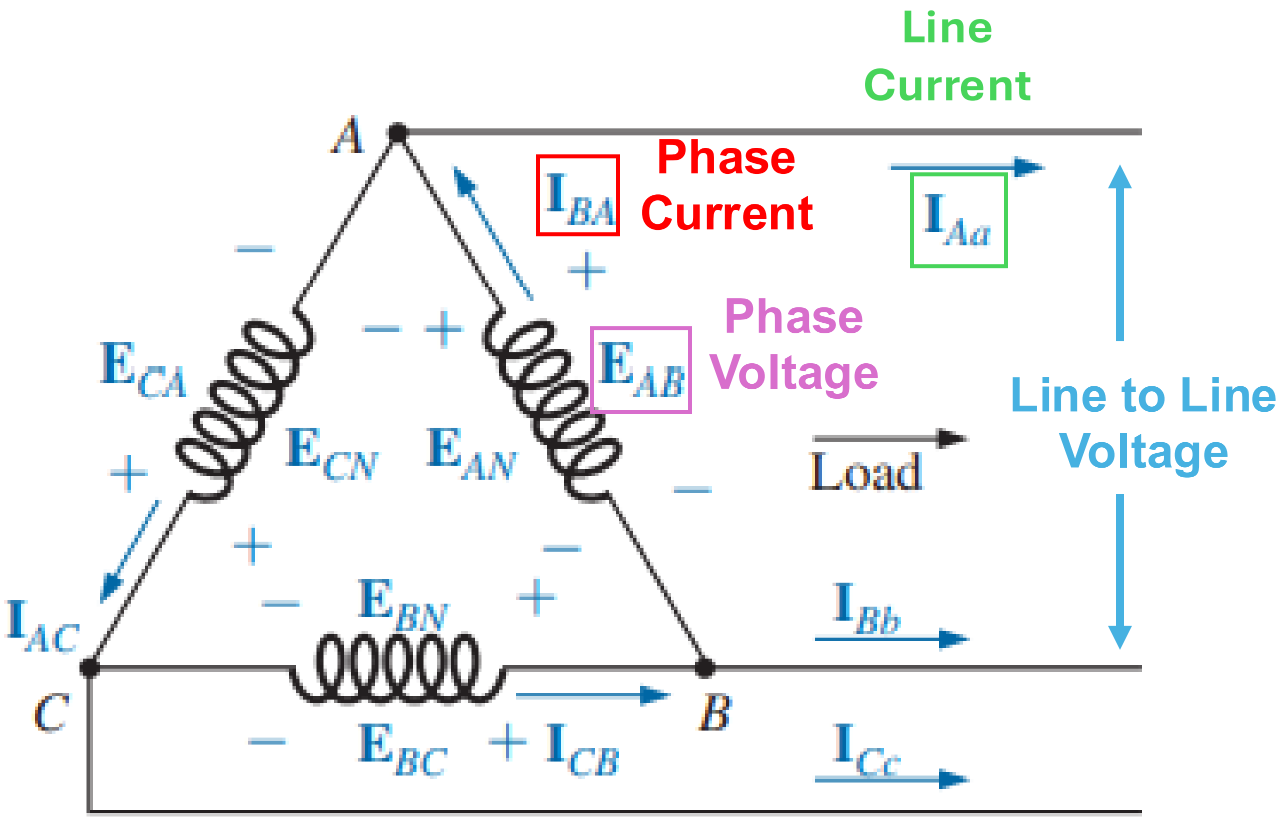


FIG. 24.14
Δ-connected generator.

Three-Phase System

Wye-Connected System

• For the Y-connected system, the line current equals the phase current for each phase.

$$\boxed{\mathbf{I}_L = \mathbf{I}_{\phi g}}$$

• Need to calculate the relation between the line voltage and the phase voltage.

Delta-Connected System

• In this system, the phase and line voltages are equivalent and equal to the voltage induced across each coil of the generator.

$$\boxed{\mathbf{E}_L = \mathbf{E}_{\phi g}}$$

• Need to calculate the relation between the line current and the phase current..

Line Voltage for Wye System

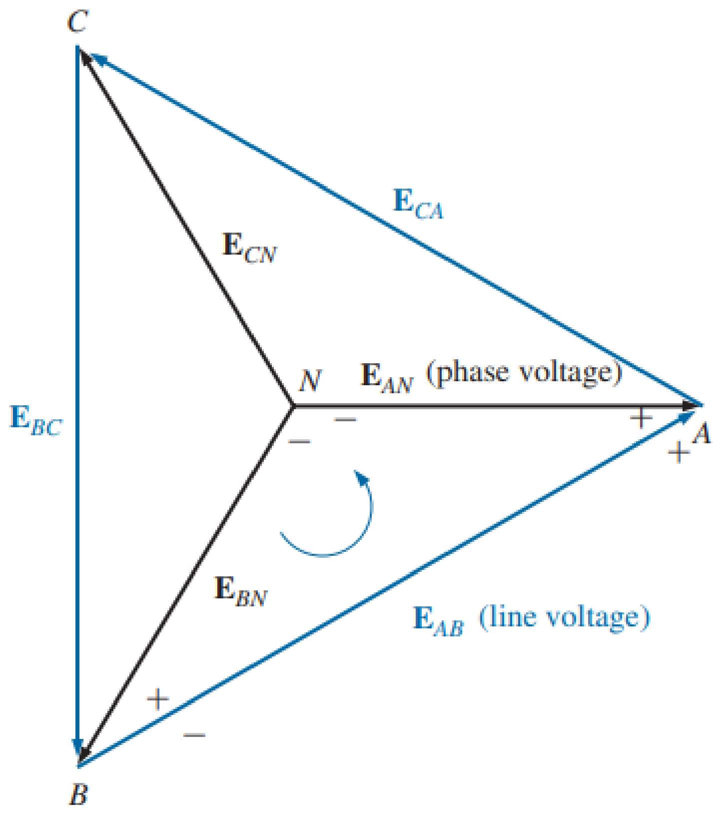


FIG. 24.5
Line and phase voltages of the Y-connected three-phase generator.

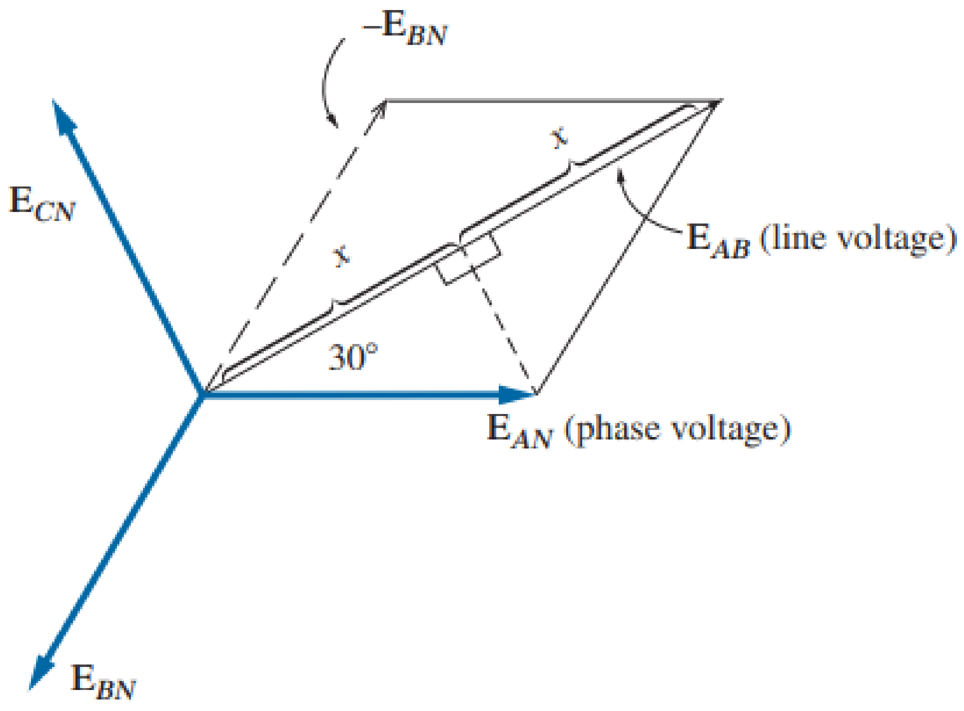


FIG. 24.6
Determining the relationship between line and phase voltages.

Line Voltage for Wye System

Phase Voltage

$$\mathbf{E}_{AN} = E_{AN} \angle 0^\circ$$
$$\mathbf{E}_{BN} = E_{BN} \angle -120^\circ$$
$$\mathbf{E}_{CN} = E_{CN} \angle +120^\circ$$

Line to Line Voltage

$$\mathbf{E}_{AB} = \sqrt{3}\,E_{AN} \angle 30^\circ$$
$$\mathbf{E}_{CA} = \sqrt{3}E_{CN} \angle 150^\circ$$
$$\mathbf{E}_{BC} = \sqrt{3}E_{BN} \angle 270^\circ$$

$$E_L = \sqrt{3}E_\phi$$

In words, the magnitude of the line voltage of a balanced Y-connected generator is 3 times the phase voltage with the phase angle between any line voltage and the nearest phase voltage at 30°.

Line Current for Delta System

Line Current

$$\mathbf{I}_{Aa} = \sqrt{3}I_{BA}\angle -30^\circ$$
$$\mathbf{I}_{Bb} = \sqrt{3}I_{CB}\angle -150^\circ$$
$$\mathbf{I}_{Cc} = \sqrt{3}I_{AC}\angle 90^\circ$$

$$\boxed{I_L = \sqrt{3}I_{\phi g}}$$

Using the same procedure to find the line current as was used to find the line voltage of a Y-connected generator produces the following: with the phase angle between a line current and the nearest phase current at 30°.

Phase Sequence

- The phase sequence may also be regarded as the order in which the phase voltages reach their peak (or maximum) values with respect to time.

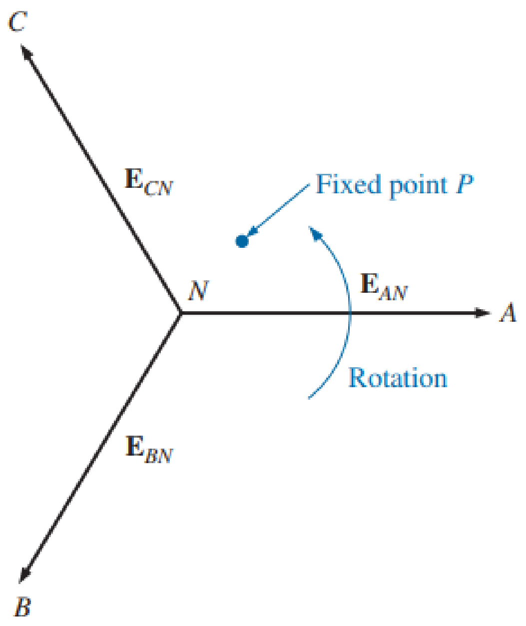


FIG. 24.7
Determining the phase sequence from the phase voltages of a three-phase generator.

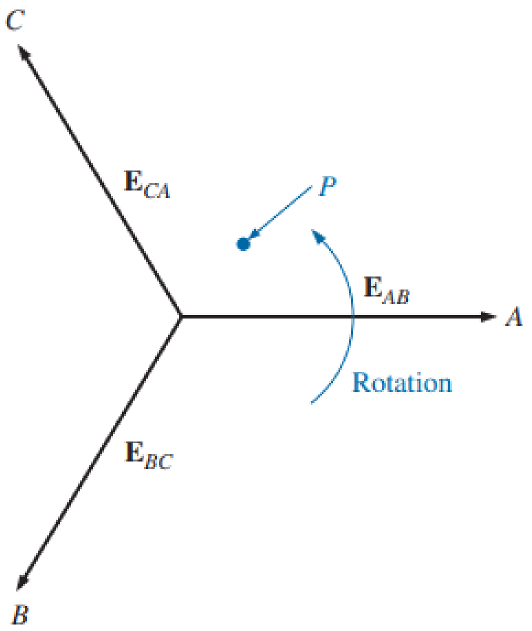
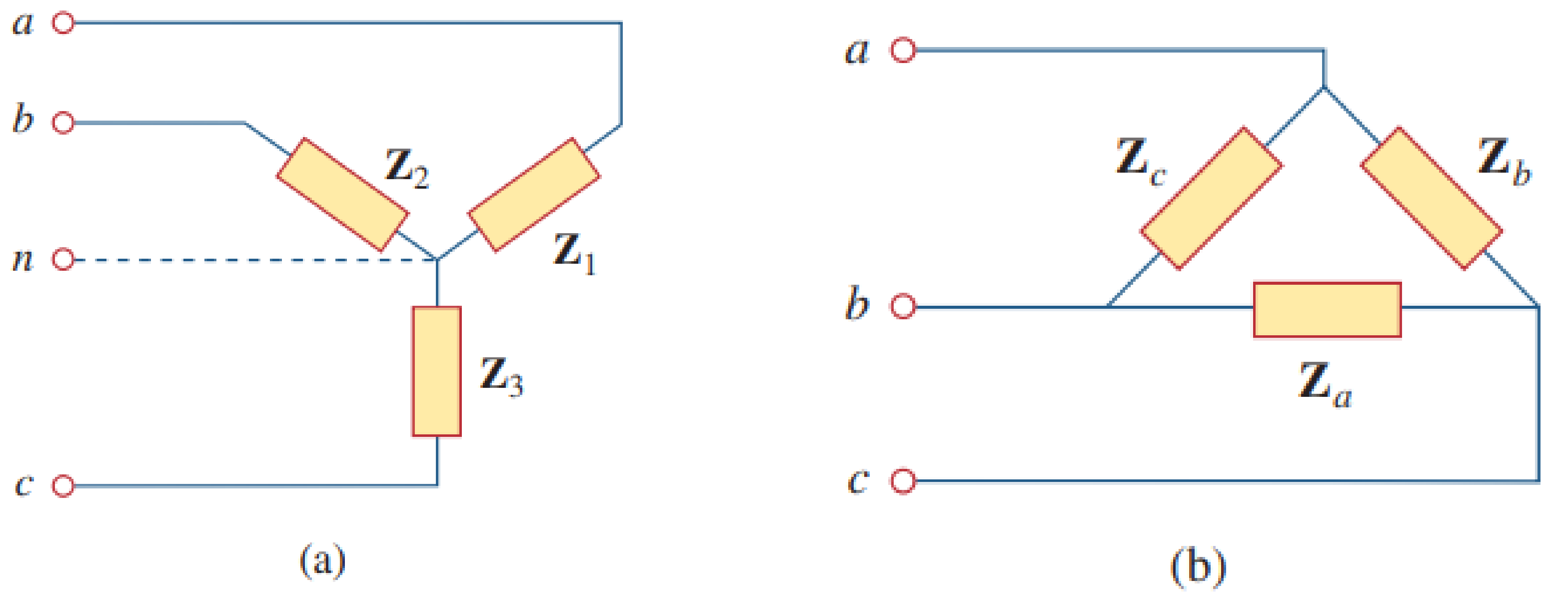


FIG. 24.8
Determining the phase sequence from the line voltages of a three-phase generator.

Balanced System

- **Balanced Phase Voltage:** Balanced phase voltages are equal in magnitude and are out of phase with each other by 120.
- **Balanced Load:** A balanced load is one in which the phase impedances are equal in magnitude and in phase.



Two possible three-phase load configurations: (a) a Y-connected load, (b) a delta-connected load.

Three-Phase System

- Since both the three-phase source and the three-phase load can be either wye- or delta-connected, we have four possible connections:

  a) Wye-Wye connection (i.e., Wye-connected source with a Wye-connected load).
  b) Wye-Delta connection.
  c) Delta-Delta connection.
  d) Delta-Wye connection

Example #1

EXAMPLE 24.1 The phase sequence of the Y-connected generator in Fig. 24.11 is *ABC*.

a. Find the phase angles θ_2 and θ_3.
b. Find the magnitude of the line voltages.
c. Find the line currents.
d. Verify that, since the load is balanced, $\mathbf{I}_N = 0$.

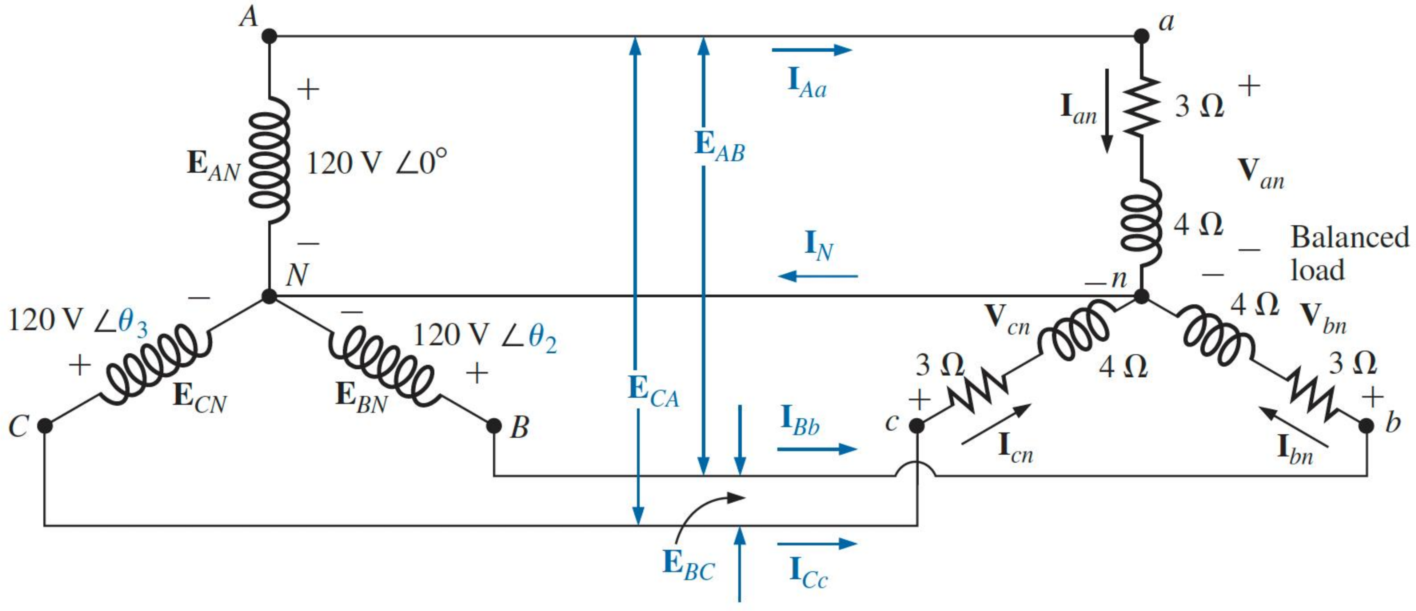


FIG. 24.11
Example 24.1.

Example #2

EXAMPLE 24.2 For the three-phase system in Fig. 24.13:

a. Find the phase angles θ_2 and θ_3.
b. Find the current in each phase of the load.
c. Find the magnitude of the line currents.

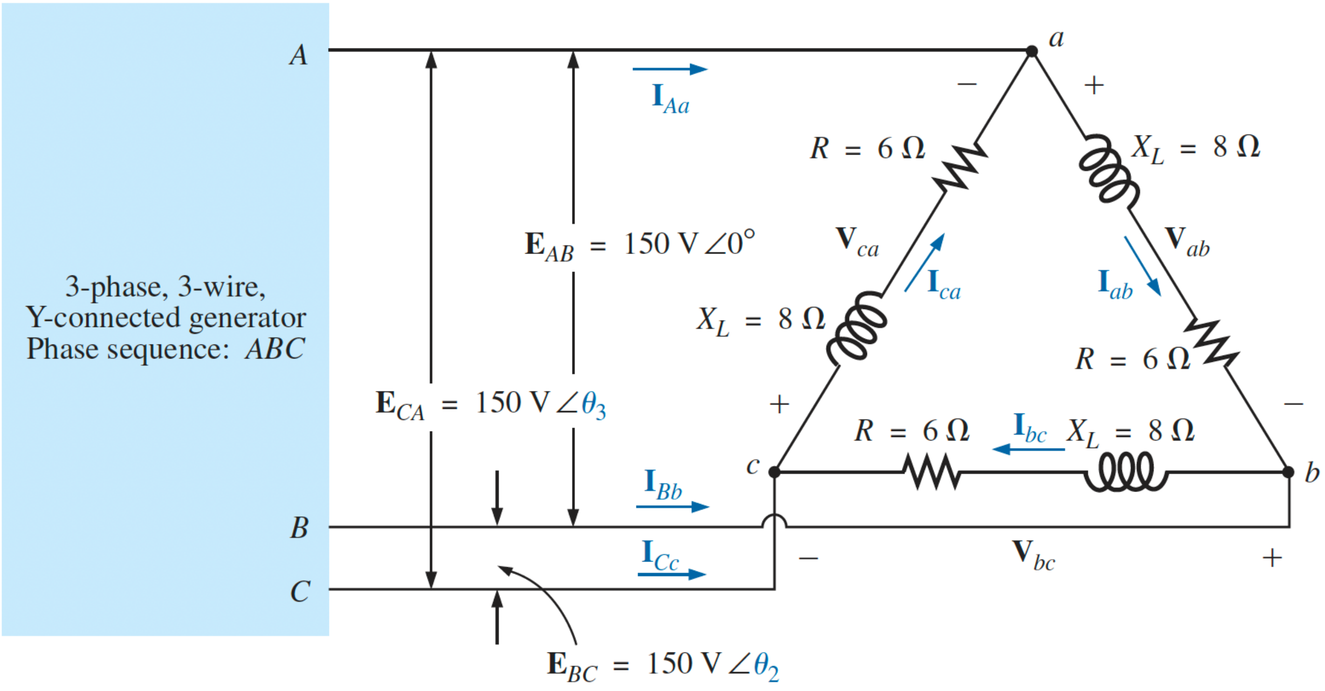


FIG. 24.13
Example 24.2.

Example #3

EXAMPLE 24.3 For the Δ-Δ system shown in Fig. 24.16:

a. Find the phase angles θ_2 and θ_3 for the specified phase sequence.
b. Find the current in each phase of the load.
c. Find the magnitude of the line currents.

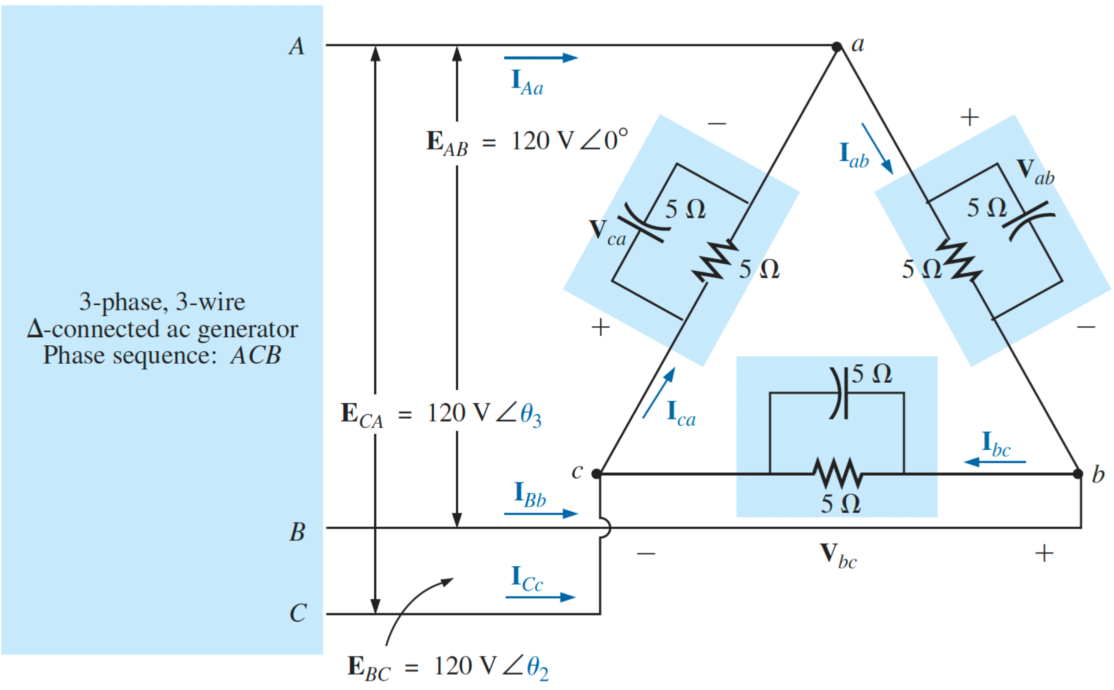


FIG. 24.16
Example 24.3: Δ-Δ system.

Example #4

EXAMPLE 24.4 For the Δ-Y system shown in Fig. 24.17:

a. Find the voltage across each phase of the load.
b. Find the magnitude of the line voltages.

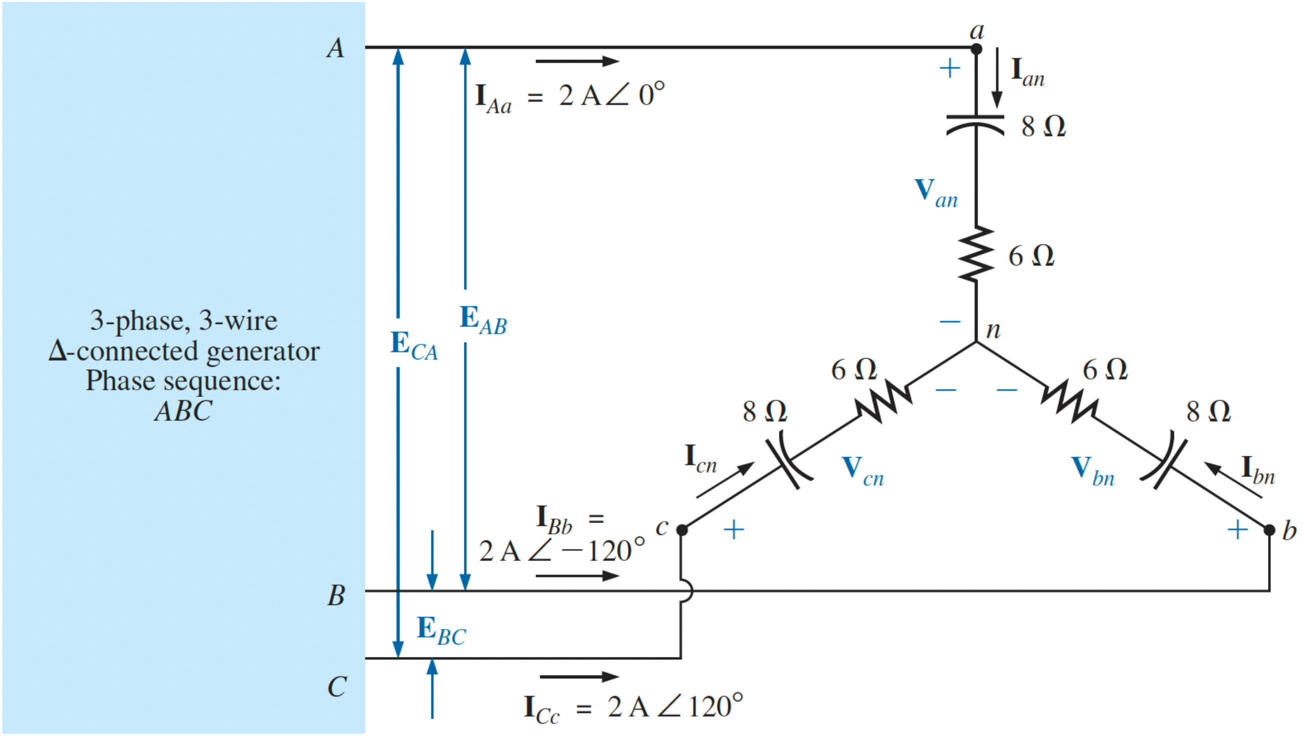


FIG. 24.17
Example 24.4: Δ-Y system.

POWER: Wye/Delta-Connected Balanced Load

Average Power

Single Phase

$$P_\phi = V_\phi I_\phi \cos \theta_{I_\phi}^{V_\phi} = I_\phi^2 R_\phi = \frac{V_R^2}{R_\phi} \quad \text{(watts, W)}$$

Three Phase

$$P_T = 3P_\phi \quad \text{(W)}$$

$$P_T = \sqrt{3} E_L I_L \cos \theta_{I_\phi}^{V_\phi} = 3 I_L^2 R_\phi \quad \text{(W)}$$

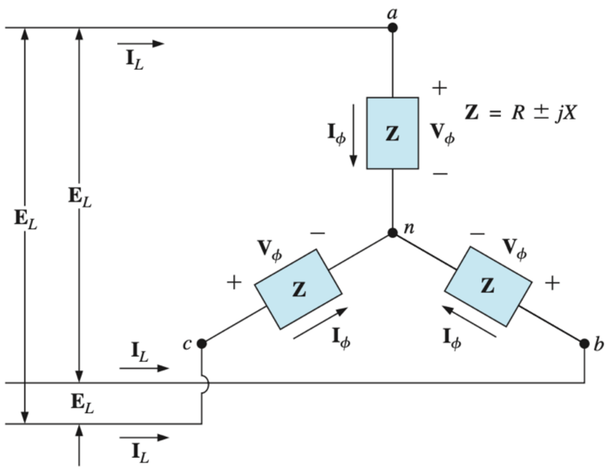


FIG. 24.18
Y-connected balanced load.

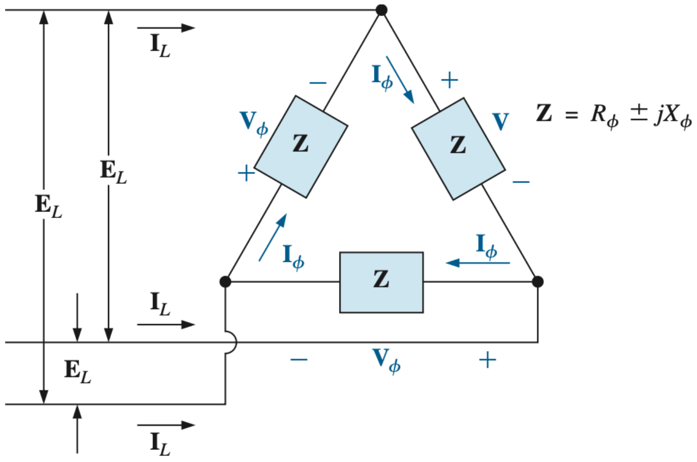


FIG. 24.20
Δ-connected balanced load.

POWER: Wye/Delta-Connected Balanced Load

Reactive Power

Single Phase

$$Q_\phi = V_\phi I_\phi \sin\theta^{V_\phi}_{I_\phi} = I_\phi^2 X_\phi = \frac{V_X^2}{X_\phi} \quad \text{(VAR)}$$

Three Phase

$$Q_T = 3Q_\phi \quad \text{(VAR)}$$

$$Q_T = \sqrt{3}E_L I_L \sin\theta^{V_\phi}_{I_\phi} = 3I_L^2 X_\phi \quad \text{(VAR)}$$

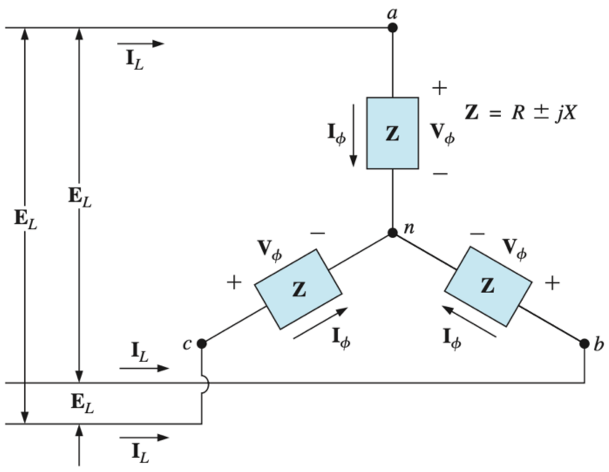


FIG. 24.18
Y-connected balanced load.

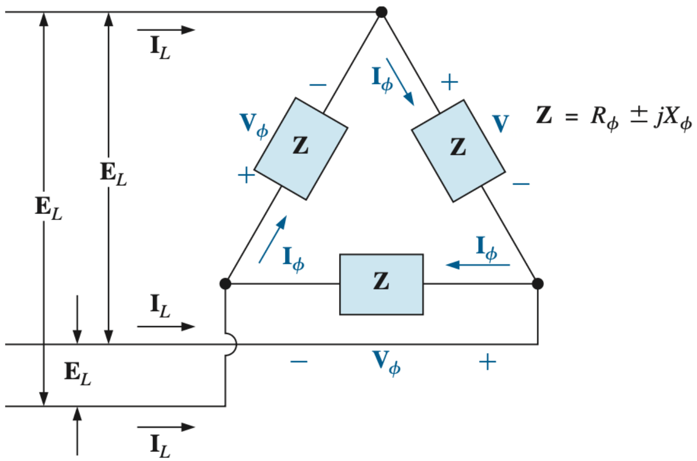


FIG. 24.20
Δ-connected balanced load.

POWER: Wye/Delta-Connected Balanced Load

Apparent Power

Single Phase

$$\boxed{S_\phi = V_\phi I_\phi} \qquad (\text{VA})$$

Three Phase

$$\boxed{S_T = 3S_\phi} \qquad (\text{VA})$$

$$\boxed{S_T = \sqrt{3}E_L I_L} \qquad (\text{VA})$$

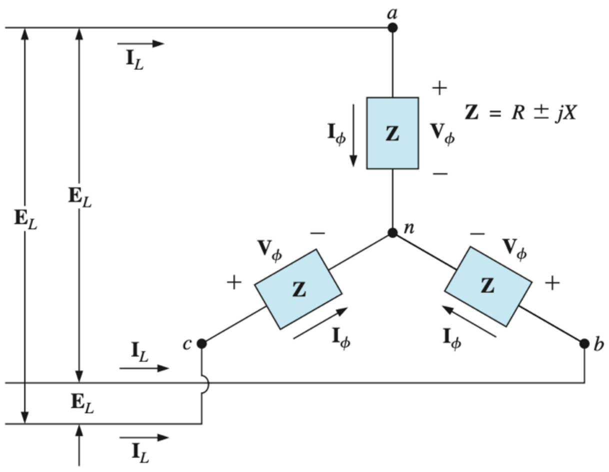


FIG. 24.18
Y-connected balanced load.

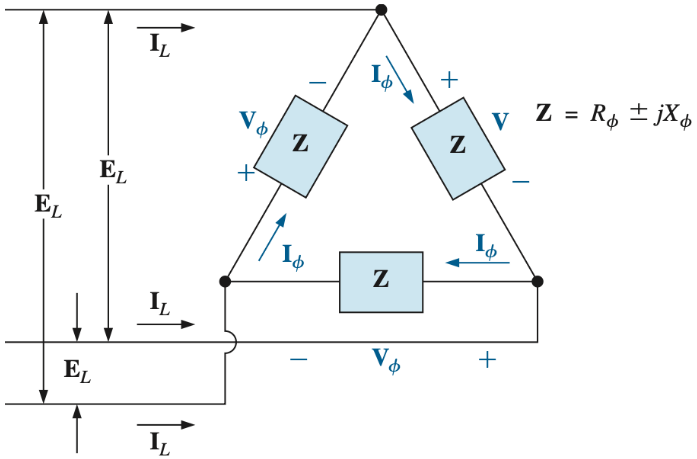


FIG. 24.20
Δ-connected balanced load.

POWER: Wye/Delta-Connected Balanced Load

Power Factor

$$\boxed{F_p = \frac{P_T}{S_T} = \cos\theta_{I_\phi}^{V_\phi}}$$

(leading or lagging)

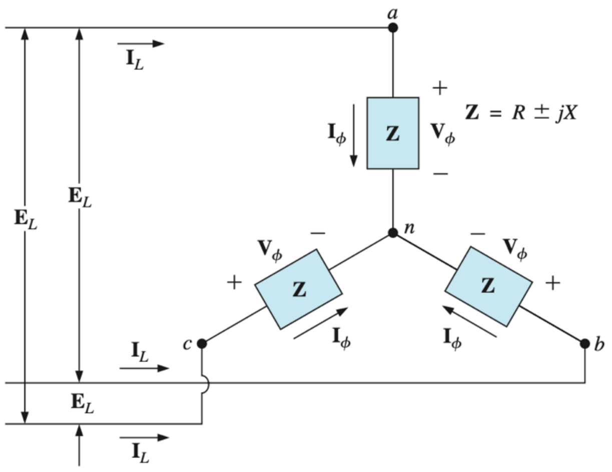


FIG. 24.18
Y-connected balanced load.

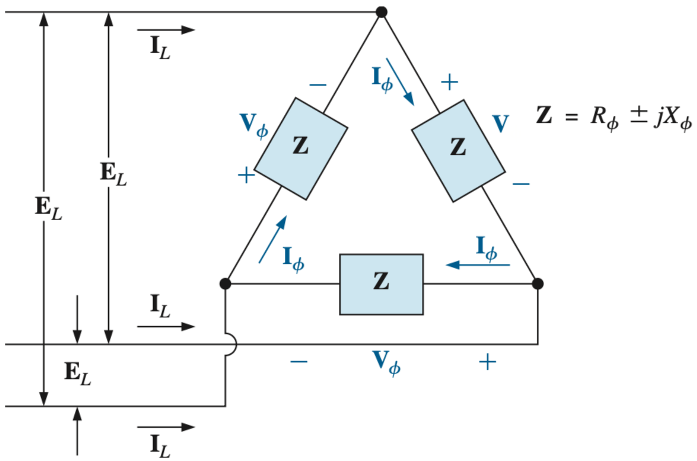


FIG. 24.20
Δ-connected balanced load.

Example #5

EXAMPLE 24.6 For the Δ-Y connected load in Fig. 24.21, find the total average, reactive, and apparent power. In addition, find the power factor of the load.

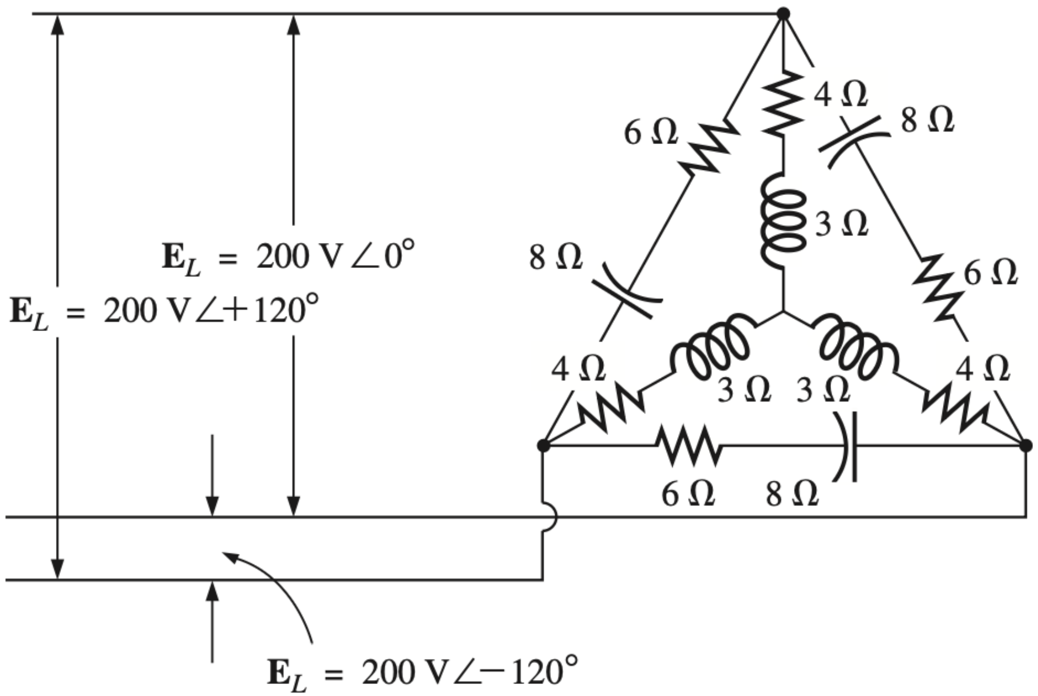


FIG. 24.21
Example 24.6.

Example #6

EXAMPLE 24.7 Each transmission line of the three-wire, three-phase system in Fig. 24.22 has an impedance of $15\ \Omega + j20\ \Omega$. The system delivers a total power of 160 kW at 12,000 V to a balanced three-phase load with a lagging power factor of 0.86.

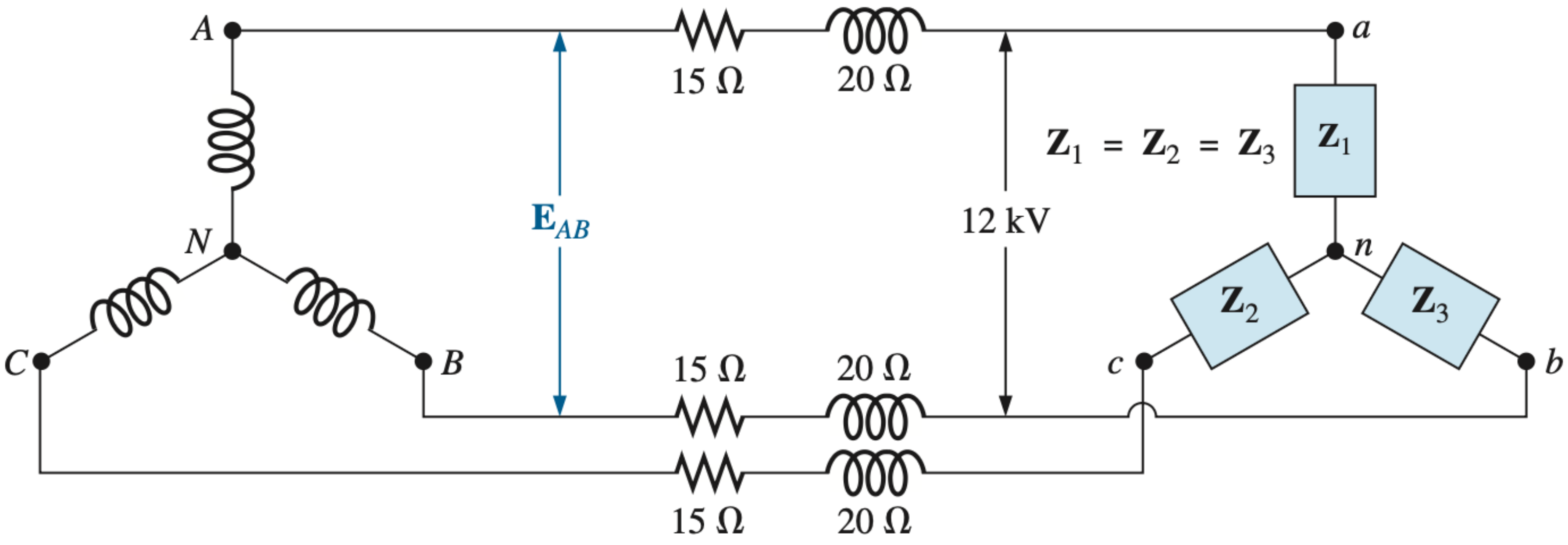


FIG. 24.22
Example 24.7.

Unbalanced Three-Phase Systems

- An unbalanced system is due to unbalanced voltage sources or an unbalanced load.
- To simplify analysis, we will assume balanced source voltages, but an unbalanced load.
- Unbalanced three-phase systems are solved by direct application of mesh and nodal analysis.

Example #7

Find the line currents in the unbalanced three-phase circuit of Fig. 12.26 and the real power absorbed by the load.

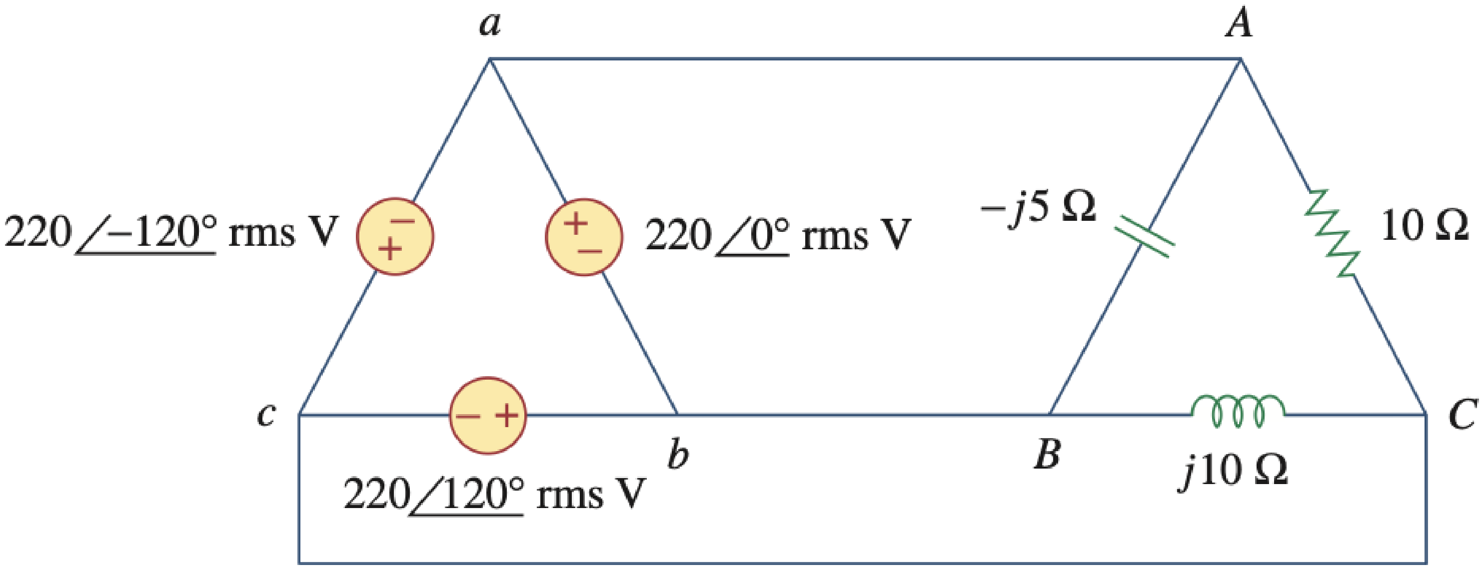


Figure 12.26
For Practice Prob. 12.10.

Example #8

For the unbalanced circuit in Fig. 12.25, find: (a) the line currents, (b) the total complex power absorbed by the load, and (c) the total complex power absorbed by the source.

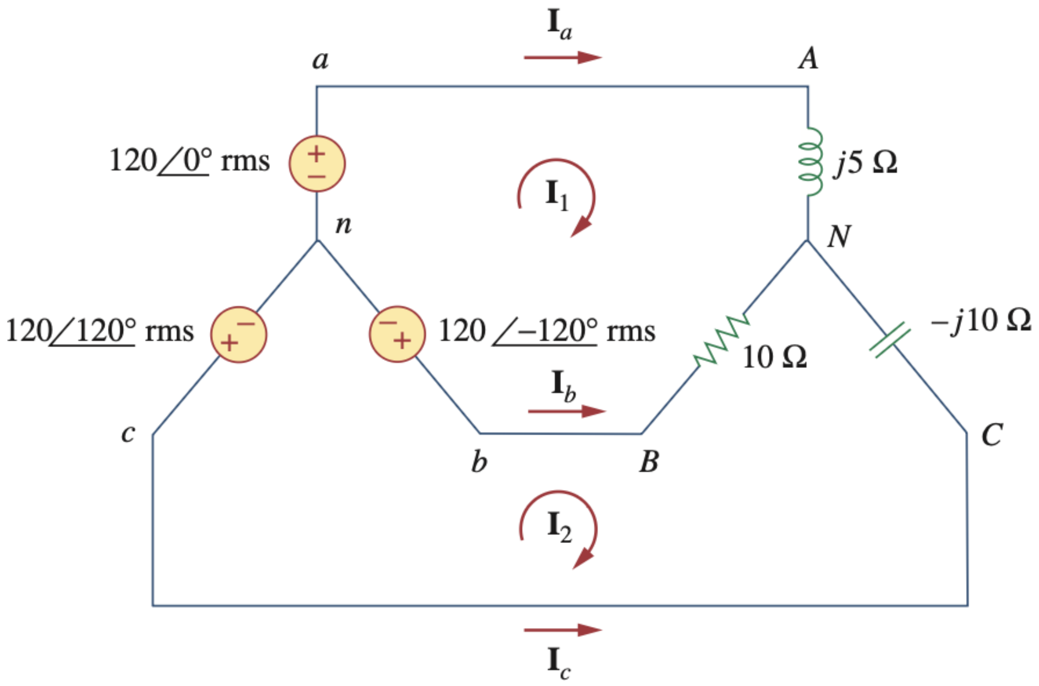


Figure 12.25
For Example 12.10.